

Data Privacy in Cloud Computing: Facebook as a Web Application

An Engineering Ethics Case Study

Introduction

- Overview of cloud computing and web applications.
- Importance of data privacy in digital platforms.
- Case study focus: Facebook.

What is Cloud Computing?

- On-demand delivery of computing services over the internet.
- Includes servers, storage, databases, networking, software.
- Used by platforms like Facebook for scalability.

Understanding Web Applications

- Web apps run on cloud infrastructure.
- Accessible via browsers without downloading software.
- Facebook is one of the most popular web apps.

How Facebook Uses Cloud Computing

- Stores and processes user data in data centers worldwide.
- Uses distributed systems for real-time interactions.
- Relies on cloud for scalability and uptime.

What is Data Privacy?

- Protecting personal information from unauthorized access.
- Ensures control, consent, and confidentiality.
- Key concern for cloud-based services like Facebook.

Facebook's Data Collection Practices

- Collects personal data: profile info, location, preferences.
- Tracks user behavior both on and off-platform.
- Uses data for ad targeting and algorithm training.

Ethical Concerns

- Informed consent: Are users aware of data use?
- Transparency: Are policies understandable?
- Monetization: Is it ethical to sell user data?

The Cambridge Analytica Scandal

- Facebook data was misused for political advertising.
- Exposed 87 million users' data without consent.
- Major turning point in privacy ethics for web apps.

Cloud Storage and User Data

- User data stored in global cloud infrastructure.
- Challenges: Cross-border data regulation.
- Questions around jurisdiction and control.

Surveillance and Privacy Intrusion

- Governments accessing Facebook data.
- Backdoors for national security purposes.
- Erosion of individual privacy rights.

User Control and Data Portability

- Facebook allows limited control over data.
- Data portability under GDPR and similar laws.
- Need for improved user empowerment.

Legal and Regulatory Landscape

- GDPR (EU), CCPA (California), and others.
- Facebook's compliance history and fines.
- Ongoing legal battles over data misuse.

Facebook's Privacy Policies

- Lengthy and complex terms of service.
- Frequent updates with little user input.
- Are they ethical and fair?

Technical Safeguards

- Encryption, authentication, and data minimization.
- Limitations in preventing internal misuse.
- Engineering challenges and responsibilities.

Facebook's Ethical Responsibilities

- Design with privacy by default.
- Provide meaningful consent mechanisms.
- Transparent reporting on data practices.

Corporate Accountability

- How should Facebook be held accountable?
- Internal ethics review boards.
- External audits and user advocacy.

User Awareness and Education

- Educate users on privacy settings.
- Raise awareness about digital footprints.
- Empower users with privacy tools.

The Role of Engineers

- Develop systems with ethical considerations.
- Balance business goals with user rights.
- Report unethical practices internally.

Future of Privacy in Cloud-Based Web Apps

- Privacy-preserving technologies: zero-trust, homomorphic encryption.
- Greater regulation and user demand for control.
- Ethics must guide future development.

Conclusion

- Facebook's case reveals key data privacy challenges.
- Ethical engineering and strong regulations are essential.
- Trust, transparency, and user control must be prioritized.